

Programming Lab 1 (CS21201)

M.Tech. I Semester (2026-27)

Assignment:- 01

Question 1: Find All Unique Combinations

Given an array of positive integers `arr[]` and a target sum `x`, write a program to find all unique combinations of elements from `arr[]` whose sum is equal to `x`.

The same element may be selected unlimited number of times. The elements within each combination must be printed in non-decreasing order. The combinations themselves should be printed in ascending order, i.e., the combination having the smallest first element should appear first.

If no combination is possible, print "Empty".

Example:

Input:

`arr[] = {2, 4, 6, 8}`

`x = 8`

Output:

`[2, 2, 2, 2]`

`[2, 2, 4]`

`[2, 6]`

`[4, 4]`

`[8]`

Question 2: Array Rotation

Write a function `rotate(arr[], d, n)` that rotates an array `arr[]` of size `n` to the left by `d` positions.

Example:

Input:

`arr[] = {1, 2, 3, 4, 5, 6, 7, 8}`

`d = 2`

Output:

`{3, 4, 5, 6, 7, 8, 1, 2}`

Question 3: Stack Implementation Using Linked List

Implement a Stack data structure using a linked list.

The following operations must be implemented:

1. size() - Returns the number of elements currently present in the stack.
2. is_empty() - Checks whether the stack is empty.
3. push() - Inserts an element at the top of the stack.
4. top() - Returns the top element without removing it.
5. pop() - Removes and returns the top element from the stack.

Demonstrate all the above operations through a suitable menu-driven program.

Question 4: Operations on Relations

Given a relation matrix representing a relation on a finite set, write programs to perform the following operations.

(a) Check Reflexivity

Write a program to determine whether the given relation is reflexive or not.

A relation R on a set A is reflexive if:

$$(a, a) \in R \text{ for every } a \in A$$

Input:

A relation matrix representing the relation.

Output:

Display whether the relation is "Reflexive" or "Not Reflexive".

(b) Check Symmetry

Write a program to determine whether the given relation is symmetric or not.

A relation R on a set A is symmetric if:

$$(a, b) \in R \Rightarrow (b, a) \in R$$

Input:

A relation matrix representing the relation.

Output:

Display whether the relation is "Symmetric" or "Not Symmetric".